Siluni Keerthiratne

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About Me

Machine Learning Engineer and First Class BSc graduate in AI and Data Science, with research experience in multimodal and multilingual NLP, vision-language models, and the evaluation of LLM-based systems. Experienced in taking ML and agentic systems from problem definition to production, and now seeking to pursue doctoral research.

Education

Robert Gordon University, Aberdeen, Scotland September 2022 - April 2026

BSc Artificial Intelligence and Data Science (First Class Honours)

Musaeus College, Colombo 7, Sri Lanka 2006 - 2022

GCE Advanced Level Mathematics

Research

Final Year Thesis - Robert Gordon University

Benchmarking and Adapting Compact Multimodal Models for Sinhala Visual Question Answering
Supervisor: Dr Ruvan Weerasinghe, September 2025 - April 2026

- Addressed the lack of datasets, benchmarks and adaptation studies for Sinhala (∼16M native speakers) vision-language models by building a 37,318-pair Sinhala VQA dataset, machine-translated from a vocabulary-stratified Visual Genome subset using Gemini Flash.
- Designed a reproducible evaluation framework combining BLEU, chrF, BERTScore, script-quality metrics and a dual LLM-as-judge protocol with length-bias penalisation, and used it to benchmark seven compact open-weight vision-language models under zero-shot conditions.
- Adapted the strongest model (Gemma-3-4B-IT) with QLoRA: fine-tuning on only 5,000 samples raised BLEU from 3.77 to 32.98 and eliminated all zero-shot output pathologies, with further gains from scaling to 33,000 samples without saturation.
- Showed that sequential continual pre-training on the MADLAD-400 Sinhala corpus degraded performance relative to direct fine-tuning, that mixed simultaneous training partially recovered the loss, and that few-shot prompting stayed well below all fine-tuned models.
- **Tools Used**: Python, PyTorch, QLoRA (parameter-efficient fine-tuning), Gemma-3, Visual Genome, MADLAD-400, LLM-as-judge.

Publications

Visual Question Answering for Low-Resource Languages: A Survey of Resources, Evaluation Methods, and Adaptation Techniques, with a Case Study on Sinhala. *Under review, Language Resources and Evaluation (Springer).*

Experience

Machine Learning Engineer, Ascentic (Pvt) Ltd. June 2026 - Present

Associate Machine Learning Engineer, Ascentic (Pvt) Ltd. July 2025 - June 2026

Software Engineer Intern, Ascentic (Pvt) Ltd. July 2024 - June 2025

National Volunteer, United Nations World Food Programme May 2022 - September 2022

Projects

Ascentic (Pvt) Ltd - Fintech Product Suite

Fintech: AI-Powered Invoice & Accounting Automation

- Owned end-to-end architecture and delivery of ML and agentic AI features for a large-scale fintech client, translating business requirements into production systems used by several Fortune 500 companies for e-invoicing and financial automation.
- Built an automated transaction-posting pipeline combining a LangGraph-based agentic workflow with per-tenant ML classifiers and a custom graph-based rule-mining engine, including an incremental/online learning loop, to classify and post financial transactions at scale.
- Built a serverless PDF-to-PEPPOL invoice extraction pipeline (AWS Lambda, Bedrock/Claude, Textract) converting unstructured PDFs and images into structured UBL e-invoice data, with OCR-based hallucination detection to validate LLM output before it reached production.
- Built a serverless multi-agent finance analyst platform (LangGraph, Bedrock/Claude) with a Text-to-SQL agent using self-correcting query refinement, stateful multi-turn sessions and SSE-streamed responses.
- Built a UNSPSC hyper-classification system using Claude-based enrichment, multilingual embeddings and vector similarity search (pgvector) over a hierarchical taxonomy to categorize financial line-item data at scale.
- Designed evaluation pipelines and an LLM-as-judge framework across the above products, validating model performance before and after deployment.
- **Tools Used**: Python, AWS Lambda, Bedrock, Textract, SQS, S3, PostgreSQL/pgvector, LangGraph/LangChain, scikit-learn, FastAPI, Serverless Framework, Docker, Git.

Ascentic (Pvt) Ltd - Partnership Product

Medical Imaging and Computer Vision Model Development

- Trained and fine-tuned YOLO-based object detection models through iterative data cleaning, feature engineering, and hyperparameter optimization workflows.
- Improved dataset quality through augmentation, labeling refinement and class balancing to improve generalization.
- Used Azure Machine Learning Studio for training and pipeline orchestration, and Comet ML for experiment tracking and comparative model evaluation.
- **Tools Used**: Python, PyTorch, pandas, NumPy, YOLOv5-YOLOv12, Azure ML, Comet ML, Git.

Skills and Certifications

ML & Research: vision-language models, QLoRA fine-tuning, multilingual NLP, LLM and agentic systems (LangGraph/LangChain), evaluation design (LLM-as-judge, custom metrics), computer vision (YOLO family).

Languages & Tools: Python, PyTorch, TensorFlow, scikit-learn, pandas, NumPy, SQL, FastAPI, .NET Core, React/Next.js, Git, Docker.

Cloud & MLOps: AWS (Lambda, Bedrock, SQS, S3), Azure Machine Learning, PostgreSQL/pgvector.

Certifications: Knowledge Graphs for RAG (DeepLearning.AI); Machine Learning in Production (DeepLearning.AI); Neo4j Fundamentals (Neo4j GraphAcademy).

References

Dr Ruvan Weerasinghe
Dean - Research, Thesis Supervisor
Informatics Institute of Technology
Colombo, Sri Lanka
Contact details available on request

Gihan Liyanage
Lead - AI & ML
Ascentic AB
Contact details available on request